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National Institute of Technology Goa

Programme Name: B.Tech
End Semester Examinations, December 2024

Course Name: **Matrices and Advanced Calculus**

Date: 09/12/2024

Duration: 3 Hours

Course Code: **MA100**

Time: 10:00 AM - 01:00 PM

Max. Marks: 100

1. Answer All Questions.
2. No marks will be given if the explanation of your answer is missing.
3. The question paper consists of **three** pages.

Part A (Matrices) ($10 \times 5 = 50$)

1. Find an upper triangular matrix A such that $A^3 = \begin{bmatrix} 8 & -57 \\ 0 & 27 \end{bmatrix}$.
2. Find the inverse of the following matrix by elementary row operations:

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}.$$

3. Express M as a linear combination of the matrices A, B, C , where $M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$, and $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$.
4. Let W be subspace of $\mathbb{R}^5$ spanned by $u_1 = (1, 2, -1, 3, 4)$, $u_2 = (2, 4, -2, 6, 8)$, $u_3 = (1, 3, 2, 2, 6)$, $u_4 = (1, 4, 5, 1, 8)$, $u_5 = (2, 7, 3, 3, 9)$. Find a subset of the vectors that form a basis of W .
5. Reduce the following matrix to an upper triangular matrix U pre-multiplying it by a sequence of elementary matrices, then find diagonal matrix D such that $A = LDU$ where

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}.$$

6. Consider the subspace U of $\mathbb{R}^4$ spanned by the vectors:

$v_1 = (1, 1, 1, 1)$, $v_2 = (1, 1, 2, 4)$, $v_3 = (1, 2, -4, -3)$ Find (a) an orthogonal basis of U ; (b) an orthonormal basis of U .

7. (a) Let λ_1 and λ_2 be two distinct eigenvalues of a matrix A and let u_1 and u_2 be eigenvectors of A corresponding to λ_1 and λ_2 , respectively. Show that $u_1 + u_2$ is not an eigenvector of A .

(b) Determine the eigenvalues and eigenvectors of $B = 2A^2 - (A/2) + 3I$ where $A = \begin{pmatrix} 8 & -4 \\ 2 & 2 \end{pmatrix}$

8. Let $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$. (a) Find all eigenvalues and corresponding eigenvectors. (b) Find a nonsingular matrix P such that $D = P^{-1} A P$ is diagonal, and P^{-1} .

9. Compute $\det A$, where

$$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

10. Let $A = 5 * \text{eye}(4) - \text{ones}(4)$ that is : $A = \begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}$.

Find the inverse and the eigenvalues and the determinant of A .

Part B (Advanced Calculus)

11. Consider the function $f(x) = \frac{x(1-x)}{1+x^2}$. [5M]

- Find the domain of $f(x)$.
- Find all horizontal, vertical asymptotes and oblique asymptote, if any
- Find all critical numbers, if any
- Find all intervals on which f is increasing and those on which f is decreasing.
- Sketch the graph of $y = f(x)$ using all of the above information. All relevant points must be labeled.

12. Suppose a function f is given along with a local linear approximation l to f at a point $P(a, b)$. Does $l(x, y) = 4y - 3x + 8$ is linear approximation of $f(x, y) = x^2y$ at $P(a, b)$? Justify your answer. [5M]

13. Suppose that a bird flies along a helical curve $x = 2 \cos t$, $y = 2 \sin t$, $z = 3t$. The bird suddenly encounters a weather front, so that the barometric pressure is varying rather wildly from point to point as $P(x, y, z) = \frac{3x^2z}{10y}$ atm. (a) Determine how the pressure is changing at $t = \frac{\pi}{4}$ minutes. (b) Using the idea of linear approximation, determine the approximate pressure at $t = \pi + 0.01$ minutes. [5M]
14. Verify the nature of the series $\sum_{k=1}^{\infty} \frac{1}{(k+1)(\ln k+1)}$. [5M]
15. Compute the volume of the solid generated by revolving the triangular region bounded by the lines $2y = x + 4$, $y = x$, and $x = 0$ about the x -axis using the washer method. [5M]
16. Find the lateral surface area of the cone generated by revolving the line segment $y = \frac{x}{2}$, where $0 \leq x \leq 4$, about the y -axis. [5M]
17. (a) Find and classify the critical points of $f(x, y) = e^{x^2-y^2}(x^2 + y^2)$. [5M]
 (b) The famous International Widget Ltd. uses aluminium, iron and magnesium to produce high-quality widgets. The quantity of widgets that may be produced using x tons of aluminium, y tons of iron and z tons of magnesium is $Q(x, y, z) = xyz$. The cost of raw materials is aluminium £ 6 per ton; iron £ 4 per ton; and magnesium £ 8 per ton. How many tons each of aluminium, iron and magnesium should be used to manufacture 1000 widgets at the lowest possible price? [5M]
18. Obtain a Fourier series for the function defined by $f(x) = \begin{cases} x, & 0 < x < \pi \\ 0, & \pi < x < 2\pi. \end{cases}$ Further, using the resulting series estimate the value of $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$ [10M]

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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, Dec-2024

Section: B

Course Name: Liberal Arts

Date: 07/12/2024

Duration: 3 Hours

Course Code: HU100

Time: 10.00 am- 1.00 pm

Max. Marks: 100

Section A: Answer in Short

12*1=12

1. Identify one folk element in Charandas Chor.
2. What is the primary conflict in Sacrifice?
3. How many kinds of plays does Bharat Muni classify?
4. Name one central theme of Tughlaq.
5. What does the term Natya mean in Natya Shastra?
6. Who tempts Charandas to break his vow of not becoming a king?
7. In which year did Tagore win the Nobel Prize?
8. Which Rasa is evoked through tragedy?
9. How does Gunavati challenge King Govinda's ideals in *Sacrifice*?
10. What is the ultimate purpose of theatre according to the *Natya Shastra*?
11. What form of theatre is *Charandas Chor* primarily based on?
12. What is Charandas's ultimate fate in the play?

13. Section B: Match the following

8*1=8

Column A	Column B
(1) Tughlaq's flawed pursuit of perfection	(A) Highlights the irony of morality in leadership
(2) Gunavati's resistance to sacrifice	(B) A reflection on the cost of political idealism
(3) Charandas's vow of honesty	(C) Challenges the hypocrisy of societal norms
(4) Symbolism of chess in Tughlaq	(D) Represents strategic decisions and moral ambiguity
(5) Conflict between faith and duty in Sacrifice	(E) Explores personal values in the face of religious doctrine
(6) Folk elements in Charandas Chor	(F) Engage audiences with humor while delivering social critique
(7) Tughlaq's transition from visionary to tyrant	(G) A cautionary tale about power corrupting ideals
(8) Charandas's ultimate sacrifice	(H) Demonstrates the power of integrity over material desires

Section C: Fill in the Blanks Questions

10*1=10

14. Tughlaq's decision to introduce _____ currency led to widespread economic chaos.
15. The character _____ impersonates a Brahmin to exploit the Sultan's policies.
16. The priest in *Sacrifice* insists on the killing of the _____ as a ritual offering.
17. *Sacrifice* is set in the kingdom of _____.
18. Charandas is executed for refusing to _____ his principles.
19. The villagers in *Charandas Chor* represent the voice of the _____ class.
20. The *Natyasastra* is considered the foundational text of _____ in Indian tradition.
21. The playhouse, as described in the *Natyasastra*, must include a _____ for the performers.
22. _____ is considered the guiding principle of dramatic composition in the *Natyasastra*.
23. The play *Tughlaq* is set in _____ century India.

Section D: Answer the following questions in not more than 200 words :

10*5=50

24. Compare the character of King Tughlaq with King Govinda from *Sacrifice* in terms of their moral and ethical dilemmas.
25. Evaluate how Habib Tanvir combines humor and tragedy in the narrative of *Charandas Chor*. How does this combination affect the audience's perception of the play's central message?
26. Evaluate the design and purpose of the playhouse as outlined in *Natya Shastra*. How does it contribute to the overall theatrical experience?
27. The *Natya Shastra* categorizes many types of plays. Critically examine how these classifications cater to diverse audience tastes and reflect cultural values.
28. In *Sacrifice*, Gunavati's steadfast devotion contrasts with King Govinda's pragmatism. Discuss how this conflict reflects larger philosophical themes.
29. Analyze the use of folk theatre techniques in *Charandas Chor* and their role in conveying moral and social critiques.
30. Critically evaluate the portrayal of idealism and disillusionment in Girish Karnad's *Tughlaq*.
31. Discuss how Habib Tanvir uses satire in *Charandas Chor* to expose corruption and moral decay in society. Provide examples of scenes where humor effectively critiques social norms.
32. Analyze the character of Charandas as an anti-hero. How does his seemingly contradictory nature (as a thief with high moral standards) challenge conventional notions of heroism?
33. Analyze the transformation of Tughlaq from a visionary ruler to a ruthless tyrant. What events and decisions contribute to this transition?

Section E: Answer any Two in detail.

2*10=20

34. Compare the portrayal of idealism and its eventual failure in Karnad's *Tughlaq*, Tagore's *Sacrifice*, and Tanvir's *Charandas Chor*. How do the protagonists' struggles reflect their internal contradictions and societal challenges?
35. Discuss the role of religion in governance and morality across the plays. How do the characters of Tughlaq, King Govinda, and Charandas use or challenge religious beliefs to assert power or maintain integrity?
36. Discuss the theme of leadership and isolation in the three plays. How do Tughlaq's alienation, King Govinda's moral struggle, and Charandas's defiance highlight the personal costs of leadership?



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Programme Name: B.Tech

End Semester Examination, December-2024

Course Name: Engineering Physics

Date: 10.12.2024

Duration: 3 Hours

Course Code: PH100

Time: 10.00 am

Max. Marks: 100

Roll no	2	4	C	S	E	10	2	7
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ANSWER ALL QUESTIONS

- Prove that the following wavefunction is satisfying the wave equation
 $\psi(x, t) = A_1 \cos k_1(x + vt) - A_2 \sin k_2(x - vt)$ (5 M)
- An electron beam is accelerated from rest through a potential difference of 200 V. (5 M)
 - Calculate the associated wavelength.
 - This beam is passed through a diffraction grating of spacing $3 \text{ \AA}$. At what angle of deviation from the incident direction will be the first maximum observed?
- An electron and a 150 gm baseball are travelling at a velocity of 220 m/s , measured to an accuracy of 0.065% . Calculate and compare uncertainty in the position of each. (6 M)
- Uncertainty in the time of an excited atom is about 10^{-8} s . What are the uncertainties in energy and in the frequency of the radiation? (5 M)
- Find the probability that a particle trapped in a box L wide can be found between $0.45 L$ and $0.55 L$ for the ground state and first excited state. (8 M)
- Show that the phase velocity of the de Broglie wave is greater than the velocity of light. Also, derive a relationship between group and phase velocity of matter waves (6 M)
- Explain the following experiments with the help of the schematic diagrams and mathematical equations. (10M)
 - Davisson-Germer Experiment
 - Photoelectric Effect
- A beam of electrons incident on the energy barrier (Tunnel) of height 0.50 eV and width of $10 \text{ \mu m}$, find the fraction of electrons transmitted through the tunnel if the energy of incident electrons is 0.40 eV (8 M)
- A particle of mass m in an infinite one-dimensional square well of width L exists in the first excited state given by: $\psi(x) = A \sin \frac{2\pi x}{L}$, (a) Find the value of A and (b) Calculate the expectation values of $\langle x \rangle, \langle x^2 \rangle, \langle p \rangle$ and $\langle p^2 \rangle$ (7 M)
- A parallel beam of light ($\lambda = 5890 \times 10^{-8} \text{ cm}$) is incident on a thin glass plate (refractive index=1.5) such that the angle of refraction into the plate is 60° . Calculate the smallest thickness of the glass plate which will appear dark by reflection. (5 M)
- Describe the construction and working of the Fabry-Perot interferometer by an oblique and normal incidence of light. (5 M)

12. A Newton's ring arrangement is used with a source emitting two wavelengths. $\lambda_1 = 6 \times 10^{-5} \text{ cm}$ and $\lambda_2 = 4.5 \times 10^{-5} \text{ cm}$ and it is found that n^{th} dark ring due to λ_1 coincides with $(n + 1)^{\text{th}}$ dark ring due to λ_2 . If the radius of curvature of the curved surface is 90 cm, find the diameter of n^{th} dark ring for λ_1 . (6 M)
13. Find the ratio of populations of the two states in a ruby laser that produces light of wavelength 6943 Å at 37 °C. (Planks constant $h = 6.62 \times 10^{-34} \text{ Js}$ and Boltzmann constant $k = 1.38 \times 10^{-23} \text{ J/K}$) (5M)
14. Explain the following with the help of the labelled diagrams with the interaction of matter and radiation in the process of (8M)
- (i) Induced absorption
 - (ii) Spontaneous Emission
 - (iii) Stimulated Emission.
- Which of the above processes is maximized for LASER operations?
15. What is total internal reflection? Explain the working principle of Optical fibers. (4M)
16. Calculate the power output of a nuclear reactor which consumes 25 gm of U-235 per day. Assume 5% reactor efficiency. The energy released per fission is 200 MeV. (3 M)
17. Explain the working of solar cells. Draw the I-V characteristics of a solar cell. Derive the expression for open circuit voltage and short circuit current of a solar cell. (4 M)

All the best



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National Institute of Technology Goa

Cuncolim, South Goa - 403 703

Name: B. Tech -I Semester (New scheme)

End Semester Examinations – Dec 2024

Course Name: Engineering Mechanics

Date: 06/12/2024

Duration: 3 hr

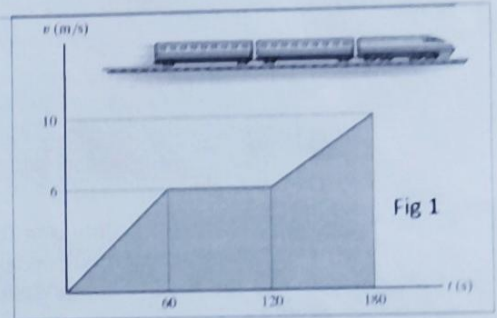
Course Code: ME100

Time: 10 am- 1 pm

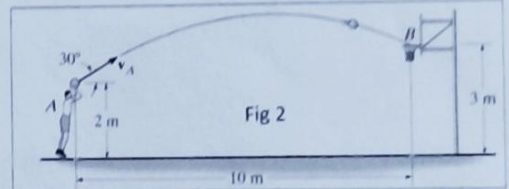
Max. Marks: 100

ANSWER ALL QUESTIONS

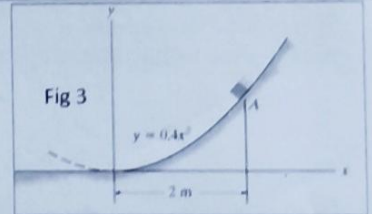
1. The $v-t$ graph for a train has been experimentally determined as shown in figure 1. From the data, construct the $s-t$ and $a-t$ graphs for the motion for $0 \leq t \leq 180$ s. When $t = 0$, $s = 0$. (8M)



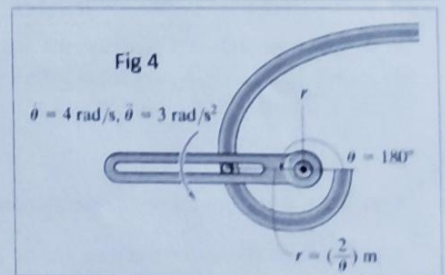
2. Neglecting the size of the ball, determine the magnitude v_A of the basketball's initial velocity and its velocity when it passes through the basket as shown in figure 2. (8M)



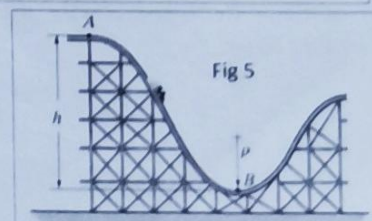
3. The box of negligible size is sliding down along a curved path as shown in figure 3 is defined by the parabola $y = 0.4x^2$. When it is at A ($x_A = 2$ m, $y_A = 1.6$ m), the speed is $v = 8$ m/s and the increase in speed is $dv/dt = 4$ m/s². Determine the magnitude of the acceleration of the box at this instant. (4M)



4. The arm is rotating at a rate of $\dot{\theta} = 4$ rad/s and $\ddot{\theta} = 3$ rad/s² and $\theta = 180^\circ$. Determine the force it must exert on the 0.5-kg smooth cylinder if it is confined to move along the slotted path as shown in figure 4. Assume the motion occurs in the horizontal plane. (8M)

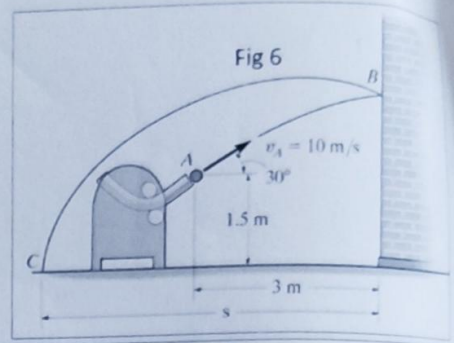


5. Determine the required height h of the roller coaster so that when it is essentially at rest at the crest of the hill A it will reach a speed of 100 km/h when it comes to the bottom B as shown in figure 5. Also, what should be the minimum radius of curvature ρ for the track at B so that the passengers do not experience a normal force greater than $4mg = (39.24m)$ N? Neglect the size of the car and passenger. (4M)



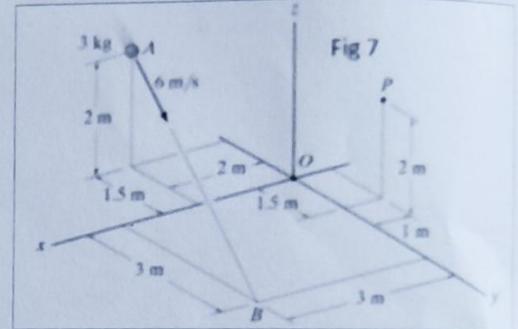
6. A pitching machine as shown in figure 6 throws the 0.5-kg ball toward the wall with an initial velocity $v_A = 10 \text{ m/s}$ as shown. Determine (a) the velocity at which it strikes the wall at B, (b) the velocity at which it rebounds from the wall if $e = 0.5$, and (c) the distance s from the wall to where it strikes the ground at C.

(10M)



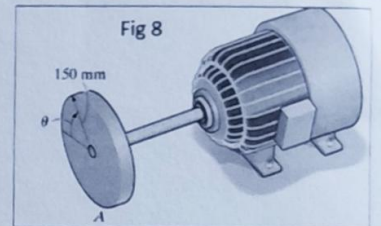
7. Determine the angular momentum H_O of the 3-kg particle at A and B about the point O as shown in figure 7.

(8M)



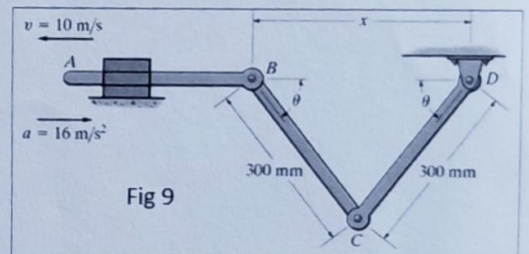
8. The motor turns the disk as shown in figure 8 with an angular velocity of $\omega = (5t^2 + 3t) \text{ rad/s}$, where t is in seconds. Determine the magnitudes of the velocity and the n and t components of acceleration of the point A on the disk when $t = 3 \text{ s}$.

(6M)



9. At the instant shown in figure 9, $\theta = 60^\circ$, and the rod AB is subjected to a deceleration of 16 m/s^2 when the velocity is 10 m/s . Determine the angular velocity and angular acceleration of link CD at this instant.

(4M)



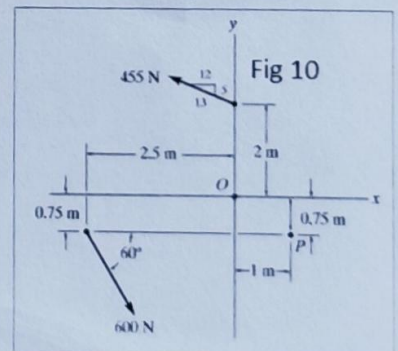
10. a) Differentiate between particle and rigid body? Give example for each.

10. b) The resultant of four forces which are acting at a point O, is along Y axis. The magnitude of forces F_1 , F_3 and F_4 are 10kN, 20kN and 40kN respectively. The angle made by 10kN, 20kN, and 40kN with X axis are 30° , 90° and 120° respectively. Find the magnitude and direction if force F_2 if the resultant is 72kN. (6+2=8M)

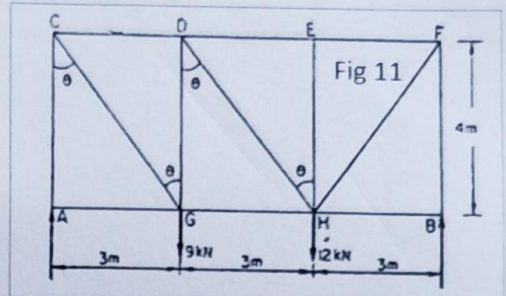
11. a) What is the condition for the system to be in equilibrium

- 11.b) Replace the force system as shown in figure 10 by an equivalent resultant force and couple moment at point O and show the result using figure.

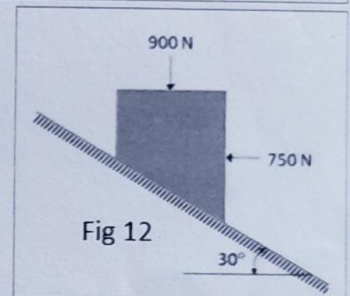
(2+7=9M)



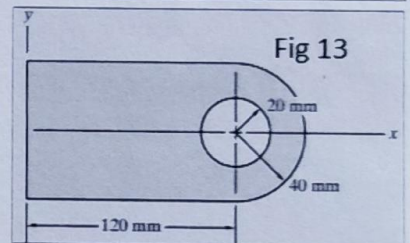
12. A truss is loaded as shown in figure 11. Find the reactions and forces in member CD, DG and GH. Also state the nature of the force. (8M)



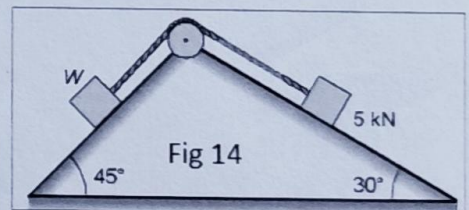
13. A block of negligible mass rests on a surface that is inclined at 30° to the horizontal plane as shown in the figure 12. When a vertical force of 900 N and a horizontal force of 750 N are applied, the block is just about to slide. Determine the coefficient of static friction between the block and surface. (4M)



14. Determine the Moment of Inertia about y-axis and Radius of Gyration for the composite area shown in the figure 13. (7M)



15. A weight of 5kN resting on a smooth surface inclined 30° to the horizontal is supported by a load of W, resting on another smooth surface inclined to horizontal by 45° as shown in figure 14. Both weights are connected with a string carried over a smooth pulley. Determine the value of weight W by using the principle of virtual work. (4M)





राष्ट्रीय प्रौद्योगिकी संस्थान गोवा
NATIONAL INSTITUTE OF TECHNOLOGY GOA
Cuncolim, Salcete, South Goa -403703
Programme Name: B.Tech.
End Semester Examinations, December-2024

Course Name: Basics of Electrical Engineering
Date: 13.12.2024
Duration: 3 hour

Course Code: EE 100
Time: 10:00 am-01:00 pm
Max. Marks: 100

ANSWER ALL QUESTIONS

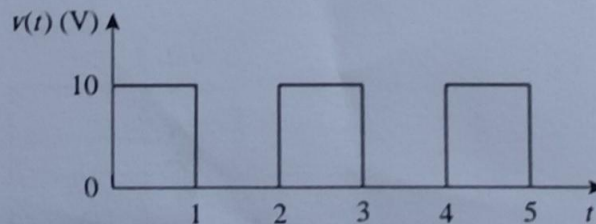
Part-A is to be answered **first** and is to be returned within **one hour** of the exam.

Part-A has to be answered in the **question paper** itself.

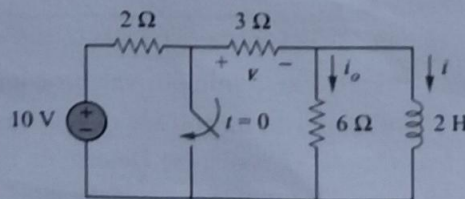
Part-B has 11 questions and carries a total of 60 marks

Part-B

1. With the help of a neat diagram, briefly explain the parts and operation of a thermal power plant. [5]
2. With the help of neat diagrams, describe the operation of a dc motor. What is the role of commutator in a dc machine [5]
3. With the help of neat sketches, briefly describe the constructional differences between a core type and a shell type transformer. [5]
4. For the periodic voltage waveform shown in Figure, determine: (a) frequency, (b) average value, (c) rms value, and (d) peak value [5]



5. The switch shown in the figure was in open position for a very long duration of time (from $t=-\infty$). At time $t=0$, the switch is closed. If v is the voltage across 3Ω resistor,



Determine: (a) v_0^- , (b) v_0^+ , (c) v_{∞} (d) Time constant of the circuit for $t \geq 0$, (e) v_i for $t \geq 0$. [6]

6. Table given below gives details of Indian electricity sector. The installed capacity of the Indian electricity sector as on year 2023 and the electricity produced in the year 2018 and 2023 are given:

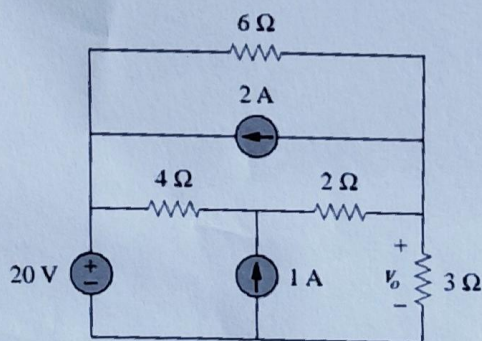
Source	Installed capacity (GW) in 2023	Electricity produced (TWh) in the year 2018	Electricity produced (TWh) in the year 2023
Coal+lignite	211.8	1022	1294
Natural gas+oil	24.8	50	32
Large Hydro	46.9	134	134
Wind	42.6	62	83
Solar	66.8	39	116
Other renewables	20.6	25	25
Nuclear	6.7	38	48
Total	416	1371	1734

- What is the percentage of electricity produced from fossil fuels in the year 2023?
- What percentage of the total installed capacity (as on year 2023) is from new & renewable energy sources (Do not include large hydro with new & renewable sources as it is a conventional source)
- Which source has shown the largest percentage growth in electricity production during the five year period (2018 to 2023). How much is this percentage growth?
- Find out the approximate capacity factor of the nuclear power plant in the year 2023.
- What percentage of the total electricity produced is from new & renewable sources (Do not consider large hydro as it is a conventional source)

[6]

7. Use superposition theorem to find voltage developed (v_o) and power dissipated across 3Ω resistor

[6]

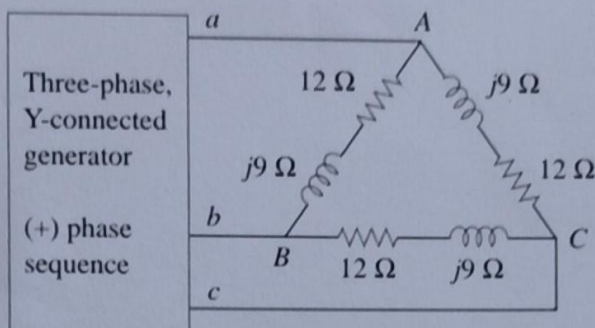


8. Design a series RLC tuning circuit. The minimum value of impedance offered by the network should be equal to 5Ω at $\omega = 1$ rad/sec. The magnitude of the impedance should be equal to 13Ω when the applied frequency becomes $\omega = 2$ rad/s. (a) Determine the value of R, L and C used. (b) What is the value of impedance offered by this network if the applied signal is dc?

[6]

9. Given that line current $I_{aA} = 40\angle -36.86^\circ$ A, The three phase source shown in the figure is balanced and n is the midpoint of the star (Y) connected generator. Determine (a) phase voltage of the source, V_{an} ; (b) phase current, I_{BC} ; (c) line voltage, V_{CA} ; and (d) real and reactive power drawn by the three phase load.

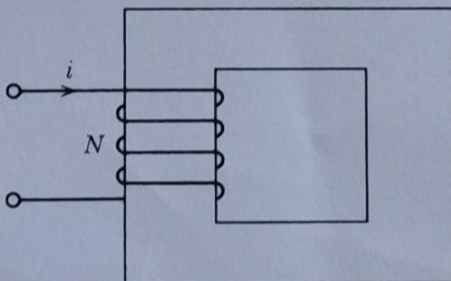
[7]



10. The mean length and the area of the core given in Figure is 200 cm and 80 cm^2 respectively. Given that the relative permeability of the core, $\mu_r = 2000$; permeability of free space, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$; number of turns, $N = 200$; and the steady current flowing through the coil, $i = 3 \text{ A}$.

- Determine the reluctance of the core
- Determine the flux in the core and its direction (clockwise/ anticlockwise)
- Determine the self-inductance of the coil
- Draw λ vs i (flux linkage vs current) curve for the given coil. Is it linear?

[7]



11. Make a rough estimate of the connected load in your class room in the given format:

[2]

Devices	No	Approx. power rating/device	Total power rating
LED bulbs			
Fans			
Projector			
Total connected load			



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, December-2024

Course Name: Computer Programming and Problem Solving

Date: 11/12/2024

Duration: 3 Hours

Course Code: CS 100

Time: 10.00 A.M. - 01.00 P.M.

Max. Marks: 100

ANSWER ALL QUESTIONS

- Q.1 a.** Evaluate the following C expression: [02 Marks]
 $A = 6 + 10 * 2 / 50 - 2 + 4$
- b.** What will be output of the following? [04 Marks]

```
#include <stdio.h>
int incr (int i)
{
    static int count = 0;
    count = count + i;
    printf("count=%d\n", count);
}

void main ( )
{
    int i, j;
    for (i = 0; i <= 4; i++)
        j = incr(i);
}
```
- c.** Provide the significance of break statement in loops. [02 Marks]
- d.** What are the different ways of initializing 1-D and 2-D array? What will happen when you access the array more than its dimension? [04 Marks]
- e.** Differentiate between structure and a union. [02 Marks]
- f.** What is a nested structure? Explain with a proper example. [04 Marks]
- g.** Will the code compile? Justify your answer. [02 Marks]

```
#include <stdio.h>
#define SWAP (a, b, c) (c t; t=a, a=b, b=t)

int main ( )
{
    int x=10, y=20;
    SWAP (x, y, int);
    printf("%d %d\n", x, y);
    return 0;
}
```
- Q.2 a.** An electricity board charges the following rates for the use of electricity: [07 Marks]
for the first 200 units 80 paise per unit: for the next 100 units 90 paise per unit: beyond 300 units Rs 1 per unit. All users are charged a minimum of Rs. 100 as meter charge. If the total amount is more than Rs 400, then an additional surcharge of 15% of total amount is charged. Write a program to read the name of the user, number of units consumed and print out the charges.
- b.** Write a C program to read N integers into an array A and to find the (i) sum of odd numbers, (ii) sum of even numbers, (iii) average of all numbers. [08 Marks]
- Q.3 a.** Define string. How string is declared and initialized? Explain string input/output functions with an example. [06 Marks]
- b.** Write a C program to sort the elements by passing array as function argument. [08 Marks]

c. Write a C program using pointers to compute the Sum, Mean and Standard deviation of all elements stored in an array of "n" real numbers. [06 Marks]

Q.4 a. Explain recursion. and write a program to find nth term of Fibonacci series. [06 Marks]

b. Give the scope and life time of the following: [08 Marks]

- 1) External variable
- 2) Static variable
- 3) Automatic variable
- 4) Register variable

~~c.~~ Write a C program to concatenate two strings without using built-in function strcat(). [06 Marks]

Q.5 ~~a.~~ Explain the difference between array and structures. [03 Marks]

~~b.~~ How many bytes in memory taken by the following C structure? [03 Marks]

```
#include <stdio.h>
```

```
struct test
```

```
{ int k;
```

```
char c;
```

```
float d[5];
```

```
};
```

```
struct test a[100];
```

~~c.~~ Determine the output of the following program: [03 Marks]

```
#include<stdio.h>
```

```
void main ()
```

```
{
```

```
union a
```

```
{
```

```
int i;
```

```
char ch[2];
```

```
};
```

```
union a u;
```

```
u.ch[0]=3;
```

```
u.ch[1]=2;
```

```
printf("%d, %d, %d\n", u.ch[0], u.ch[1], u.i);
```

```
}
```

d. What will be the output of the C program? [03 Marks]

```
#include<stdio.h>
```

```
int main ()
```

```
{
```

```
enum numbers {n1 = 1.5, n2 = 0, n3, n4, n5, n6};
```

```
printf("%d %d\n", n1, n2);
```

```
}
```

~~e.~~ Explain passing structures to a function with respect to the following: [03 Marks]

(i). Passing individual members.

(ii). Passing entire structure.

~~f.~~ Write a C program to read the details of book name, author name and price of 200 books in a library and display the total cost of the books and the book details whose price is above Rs.500. [05 Marks]

~~g.~~ Write a program to read a file and display contents with its line numbers. [05 Marks]